

AI Fundamentals Capstone Project

Workplace AI Implementation: Before and After Prompting Comparison

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Course: AI Fundamentals in the Workplace

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This project demonstrates how my use of AI improved after completing the course. I selected a real recurring workplace task and compared my previous prompting approach with a new structured approach using the R.A.C.E. framework. The project also demonstrates data masking, iterative refinement, verification, and the impact of AI on time and output quality.

Target Task:

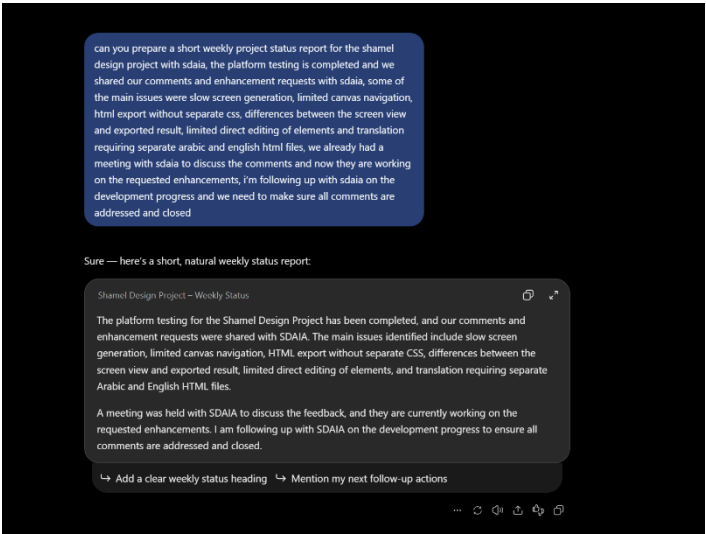
Task: Preparing a weekly project status report from project updates.

This is a recurring workplace task that involves summarising progress, identifying key issues, and presenting next steps in a clear and professional format.

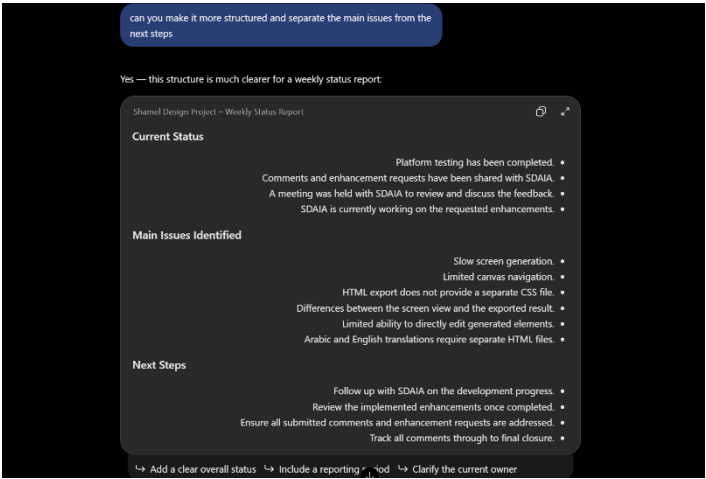
Before the Course – Previous Prompting Approach

Before the course, I typically used direct prompts and improved the output through follow-up requests without following a specific prompting framework.

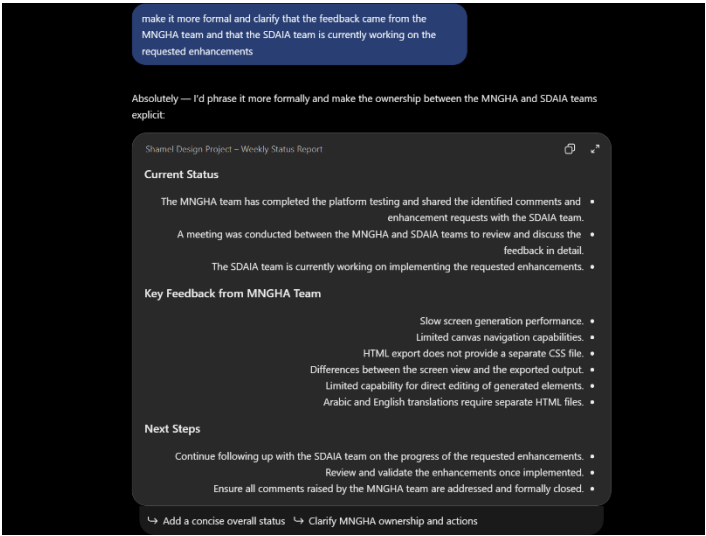
Screenshot 1 – Initial Prompt and Output



Screenshot 2 – First Follow-up Refinement



Screenshot 3 – Final Follow-up for Tone and Ownership



The final output was usable and well structured, but reaching the desired result required multiple follow-up prompts the initial prompt did not clearly define the role, audience, expected format, privacy requirements, or verification instructions.

After completing the course, the main change in my approach was to give clearer instructions from the beginning instead of waiting for the first output and then adding more details I already used to remove sensitive information before using AI, but after the course I started using data masking in a more structured way, for example by replacing names and project details with labels like [Project A] or [External Entity].

Data Masking

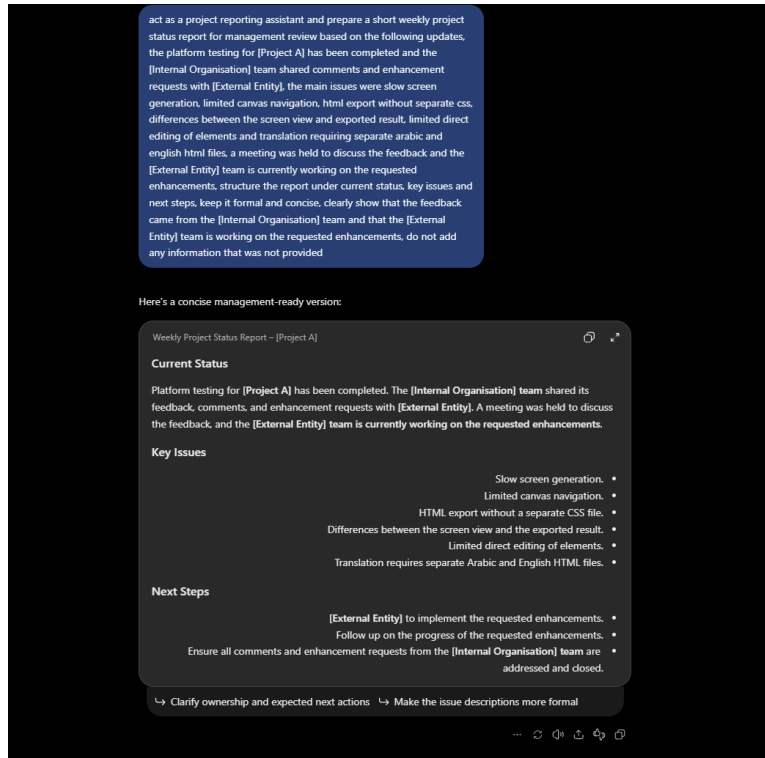
For the after-course version, I replaced identifiable project and organization details with neutral labels before using the information in the prompt.

Original	Masked
Shamel Design Project	[Project A]
MINGHA	[Internal Organization]
SDAIA	[External Entity]

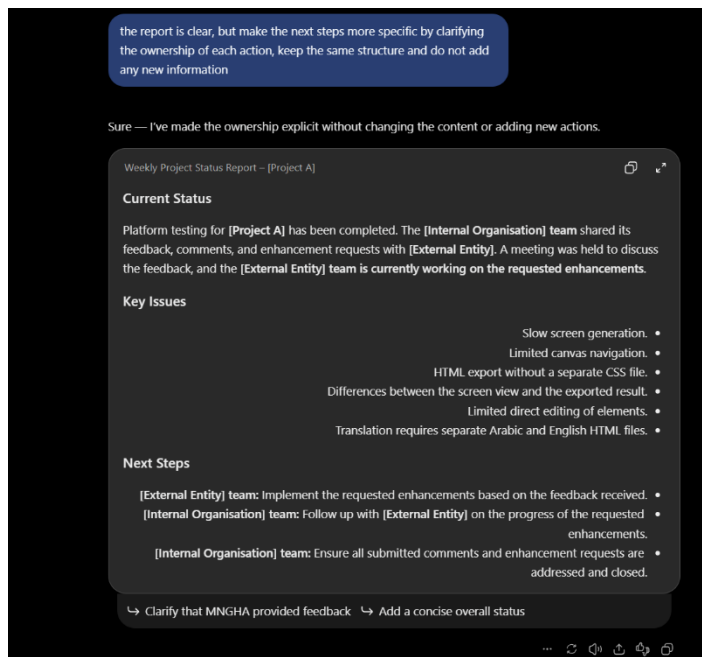
The same project updates were kept, but the identifying details were masked before creating the new prompt.

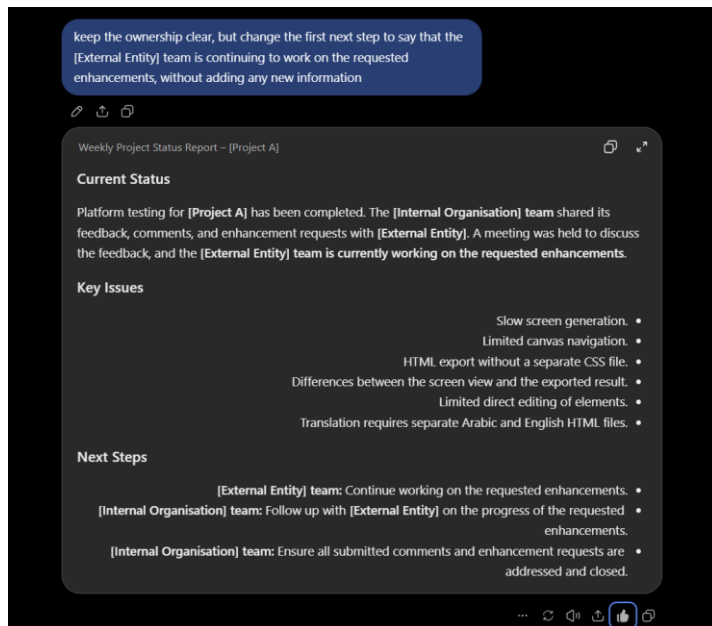
After the Course – R.A.C.E. Prompt

I used the R.A.C.E. framework to make the prompt clearer from the beginning and reduce the need for repeated follow-up edits.



Iterative Refinement





I used follow-up prompts to refine the ownership and ensure the final wording stayed aligned with the original information. After reviewing the final version, I marked the response as helpful to indicate that this wording best matched the intended format.

Verification & Impact

I reviewed the final report against the original project updates and checked that the status, key issues, ownership, and next steps were accurate. I also confirmed that no new information was added and that the masked labels were used consistently.

Time Saved

Before the course, preparing and refining the report took around 10–12 minutes because I relied on several follow-up prompts. Using the structured approach, the report took around 5–6 minutes including review and verification, saving approximately 5–6 minutes per report.

Before vs After Comparison

Before the Course	After the Course
General initial prompt	Clear structured prompt from the start
Several follow-up edits	Fewer, more targeted refinements
Sensitive details removed manually	Data masking used systematically
Structure developed after seeing the output	Structure defined in the initial prompt
Ownership clarified later	Ownership included in the prompt
Verification was informal	Final output checked against source information

The main improvement was not only the final output, but the way I controlled the process from the beginning. The structured prompt reduced unnecessary back-and-forth and made the final report easier to verify.

Final Reflection

With my academic and technical background in AI, I was already familiar with prompting and using AI tools in different tasks. What I gained from this course was a more structured and practical approach to using AI in the workplace. It helped me improve how I define requirements from the beginning, apply data masking more systematically, use targeted follow-up prompts, and verify outputs before using them. The course added a stronger workplace-focused approach to skills I already had.